Postclassical Time Maps: Theory and Interpretation

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This paper fleshes out the theory of "time maps" and demonstrates their interpretive utility. Time maps are the graphs which are produced when a narrative's fabula (i.e. the timeline of events within the story) is plotted against its syuzhet (i.e. the order in which these events are presented to readers/viewers/players). In short, time maps are a digital and data-driven means of visualizing a narrative's temporal relations. My previous work on time maps received the Paul Fortier Prize at the 2019 Digital Humanities Conference in Utrecht for best paper by an emerging scholar. An introduction to the theory of time maps, along with my methodology for producing them, is available in *Cultural Analytics* (Yeager 2023). The present paper discusses three advanced theoretical concepts: narrative velocity; non-linear narrative chronotopes; and ambiguous temporalities. I also offer a comparative analysis of Orson Welles's *Citizen Kane* and Assata Shakur's *Assata: An Autobiography*. Though these two narratives differ in terms of thematics, genre, and medium, they become curious bedfellows when their time maps are juxtaposed.

To start, time maps prompt a re-evaluation of Gérard Genette definition of narratological velocity, which he establishes by comparing " *a length* (that of the text, measured in lines and in pages)" to " *a duration* (that of the story, measured in seconds, minutes, hours, days, months, and years)" (1980: 87-8). He borrows from algebra-based physics, where velocity is the ratio between length and time, to state that narratological velocity is the ratio of syuzhet to fabula. A flaw with Genette's analogy is he is not comparing a length to a time; instead, he is comparing approximations of two different types of time: the character's and the reader's. A consequence of this confusion is that his ratio is upside down, because readers must directly engage with the narrative's syuzhet in order to access its fabula. Instead, it makes more sense to consider syuzhet as the independent variable and fabula as the dependent variable, so that a narrative's velocity is understood in terms of how the fabula fluctuates with regard to the syuzhet. When this understanding is used, the velocity of any given segment of narrative corresponds to the slope of its line on a time map. Figure 1 offers a rudimentary comparison of how this plays out with regard to Genette's five categories of pause, stretch, scene, summary, and ellipsis. In short, the line segments with a steeper slope have a higher velocity (e.g. A sentence-long summary like "she worked all day" has a steeper slope than the full text of *Mrs. Dalloway*, because the latter stretches out the day's events over a few hundred pages).

Figure 1: Comparing the five classical types of narrative velocity.

Pushing this reformulation of narrative velocity means shifting from an algebra-based understanding of velocity to a calculus-based one. Within an algebra based framework, we can only plot line segments on a time map. While line segments can represent many narrative phenomena, they cannot – for example – represent segments of syuzhet where the fabula speeds up or slows down. Consider a film clip which shows characters discussing an impending emergency, then segues into a fast-forwarded sequence of them preparing for this emergency. This storytelling device involves a narratological acceleration – i.e. a shift from one flow of narratological time to another – and we need to use a parabola to accurately represent this shift. My previous work refers to the line segments which get plotted on a time map as "narratological chronotopes;" thinking more broadly requires the incorporation of "non-linear narratological chronotopes."

The final theoretical phenomenon that I consider involves temporal ambiguities which arise when we step outside the realm of "natural" narratives. Consider *The Twilight Zone* episode, "A Kind of Stopwatch." In this story, a character finds a stopwatch which literally stops time for the rest of the world, though it proceeds at a regular pace for him. Time maps are useful for contemplating this kind of structure because there are two potential ways to visualize it: "explicitly adopting some metanarrative frame" or "selecting a privileged viewpoint" (Wittenberg 2013: 7). When using the first strategy, we portray this phenomenon as a Genettean pause, because time for the external world has stopped (Figure 2, left). When using the second strategy, we portray it as a Genettean scene, since time is still flowing for the main character (Figure 2, right). While there is no definitive way to resolve this interpretive impasse, a time map clarifies the problem and prompts a contemplation of the constructed nature of time in narrative.

Figure 2: Two contrasting time maps of “A Kind of Stopwatch.”

In addition to these theoretical contributions, I offer two interpretations which are rooted in time maps. In Genette’s terms, “we can characterize the temporal stance of a narrative only by considering at the same time all the relationships it establishes between its own temporality and that of the story it tells” (1980: 155). Time maps, in other words, are useful interpretive tools because they visualize a narrative’s temporal structure, the embodiment of its philosophical understanding of time.

Consider the time map of Orson’s Welles’s *Citizen Kane* (Figure 3). This graph approximates the shape of a triangle whose sides are the vertical blue dots on the left, the horizontal black line across top, and the multicolored line segments below. These respectively correspond to the opening newsreel, the reporter's interviews, and the events of Kane's life as recounted by the interviewees. The film famously features several narrators, whose spans of syuzhet are demarcated by vertical gray lines, yet these narrators recall Kane's life in almost exactly the order it unfolded, so the triangle’s hypotenuse is nearly linear. As a result, almost every narratological chronotope in Kane lies along the perimeter of this triangle. There is one glaring exception: when Kane is forced to sell controlling interest in his newspaper, the Inquirer , which is shown in red. On a time map, this loss literally lies at the center of the film, and thus shapes the viewer's understanding of Kane's life. The triangle's corners are similarly significant. The film opens in the top left corner when Kane speaks his last word, the bottom left corner depicts him sledding as a child, and the top right corner shows that same sled getting thrown into a furnace. Just as the vertices define a triangle, Rosebud delineates Kane's life. Commentators typically presume that Rosebud represents Kane’s inner child, a wounded and vulnerable core which he attempts to hide behind a facade of “success.” But through the geometric mapping of a time map, Kane's defining feature is indeed his failed ambition and the fixation on his lost childhood is merely a shell of nostalgia that is rooted in cognitive dissonance.

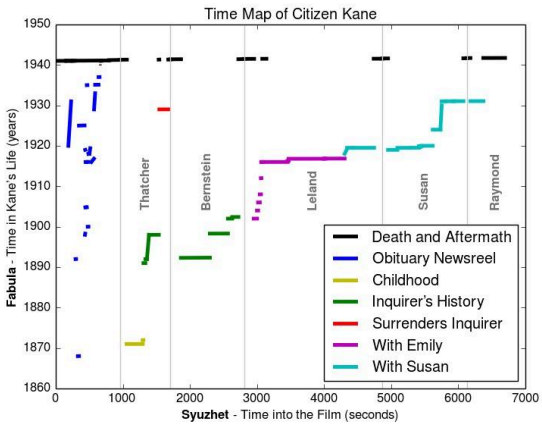

Figure 3: Time map of Citizen Kane .

Interestingly, Assata Shakur uses a very similar temporal structure to Welles in her autobiography (Figure 4), though she deploys this temporality to very different ends. The book opens with a chart which chronicles her trial dates, shown in green. This doesn’t reach all the way back to her childhood, like *Kane*’s newsreel, but it fulfils a similar purpose by giving the reader a preview of what’s to come. Shakur’s first chapter discusses the incident which led to her imprisonment, then her second chapter discusses her early childhood. From there, the chapters alternate between her experiences in prison and her life story, following a “zigzag” pattern which mirrors *Kane*’s alternation between interviews and flashbacks (Montfort 2007: 89). Yet there is a key structural difference between these narratives, which occurs once their internal timelines converge. For *Kane*, the past only catches up to the present posthumously; the end of the character’s life story roughly coincides with the end of the narrative itself. But for Shakur, her storylines instead converge at the moment of her final trial, but this is not the end of her tale, as demonstrated by the tail in the time map’s upper right corner. Instead, Shakur goes on to narrate her experiences of the trial, its resultant conviction, and eventually the details of her life after escaping from prison. The distinction between these two shapes is partially attributable to genre: the narrator of a real-world autobiography must be alive to narrate it, typically in a manner such that “self-distance must end in self-presence as the events of a life catch up with the moment of telling it” (Currie 2007: 61). Yet these theoretical constructs carry over to the narrative’s thematics: *Kane* is the pessimistic story of a rich white man who tries to make himself unknowable to others, whereas *Assata* is the hopeful story of an incarcerated Black woman who tries to imagine a more equitable future – thus the tail.

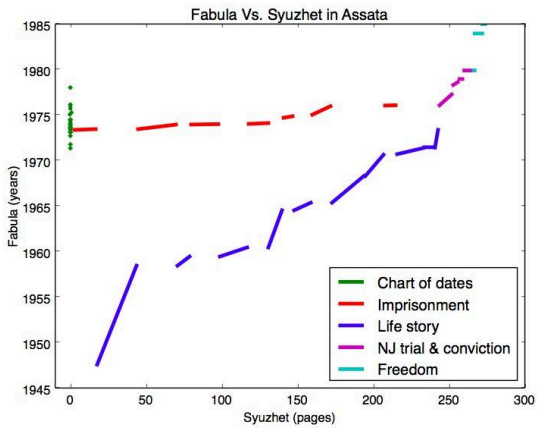

Figure 4: Time map of *Assata: An Autobiography*.

In conclusion, this paper demonstrates two key utilities of time maps. First, they can shed light on abstract narratological structures. Furthermore they offer unique ways to interpret and compare stories, based on their embodiment of temporal relations.

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